FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2022

TEST DATE: 18-12-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

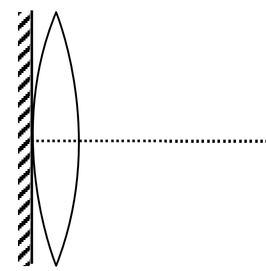
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A thin convex lens of focal length 'f' is put on a plane mirror as shown in the figure. When an object is kept at a distance 'a' from the lens mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of 'a' is



(A) 3f

(B) $\frac{3}{2}f$

(C) f

(D) 2f

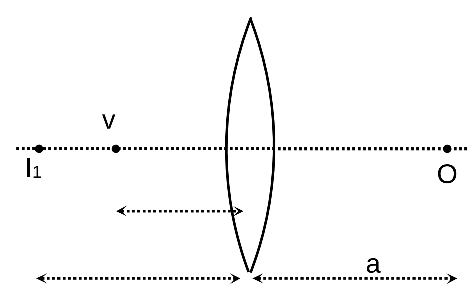
Ans. D

Sol. When object is kept at a distance 'a' from thin convex lens.
By lens formula →

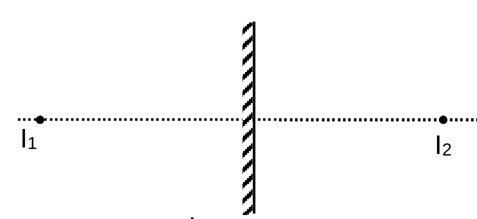
$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{(-a)}=\frac{1}{f}$$

$$\Rightarrow \frac{1}{v}=\frac{1}{f}-\frac{1}{a} \qquad \ldots(i)$$



Mirror forms image at equal distance from mirror

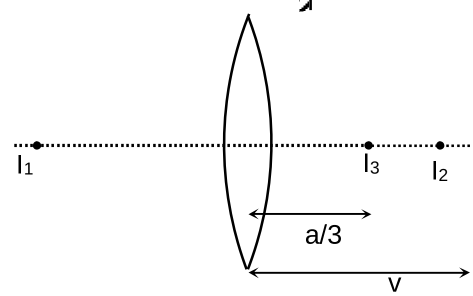


Now again from lens formula,

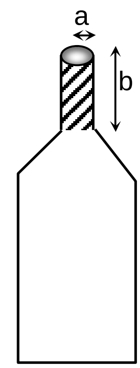
$$\frac{3}{a}-\frac{1}{v}=\frac{1}{f}$$

$$\frac{3}{a}-\frac{1}{f}+\frac{1}{a}=\frac{1}{f}$$

$\Rightarrow$ a = 2f



2. A bottle has an opening of radius a and length b. A cork of length b and radius (a + Δa) where (Δa <<a) is compressed to fit into opening completely as shown in the figure. If the bulk modulus of cork is B and frictional coefficient between the bottle and cork is μ then the force needed to push the cork into the bottle is



(A) $(\pi\mu Bb)a$

(B) $(2\pi\mu Bb)\Delta a$

(C) $(\pi\mu Bb)\Delta a$

(D) $(4\pi\mu Bb)\Delta a$

Ans. D

Sol. Stress = $\frac{\text{normal force}}{\text{Area}}$

$\Rightarrow$ stress = $\frac{N}{(2\pi a)b}$ …(i)

Stress = B $\times$ strain …(ii)

From equation (i) and (ii)

$$\frac{N}{(2\pi a)b} = B\left[\frac{2\pi a(\Delta a)\times b}{\pi a^2 b}\right]$$

$$\Rightarrow N = \frac{B(2\pi a)^2 \Delta ab^2}{\pi a^2 b} = (4\pi b)(\Delta a)$$

Force needed to push the cork

f = μN

= $\mu(4\pi b)(\Delta a)$

= $(4\pi\mu Bb)\Delta a$

3. A physical quantity $y = \frac{a^4 b^2}{(cd^4)^{1/3}}$ has four variables a, b, c and d. The percentage error in a, b, c and d are 2%, 3%, 4% and 5% respectively. The error in y will be

(A) 6%

(B) 11%

(C) 12%

(D) 22%

Ans. D

Sol. $\frac{\Delta y}{y} = 4\frac{\Delta a}{a} + \frac{2\Delta b}{b} + \frac{1}{3}\frac{\Delta c}{c} + \frac{4}{3}\frac{\Delta d}{d}$

$= 4 \times 2 + 2 \times 3 + \frac{1}{3} \times 4 + \frac{4}{3} \times 5 = 22$

4. The ratio of contributions made by the electric field and magnetic field components to the intensity of an EM wave is

(A) c : 1

(B) c^2 : 1

(C) 1 : 1

(D) $\sqrt{c} : 1$

Ans. C

Sol. Average energy by electric field E_0 is

$U_{av} = \frac{1}{2}\varepsilon_0 E_0^2$ (But $E_0 = cB_0$)

Hence, $(U_{av})_{\text{electric field}} = \frac{1}{2}\varepsilon_0 (cB_0)^2 = \frac{1}{2}\varepsilon_0 c^2 B_0^2$

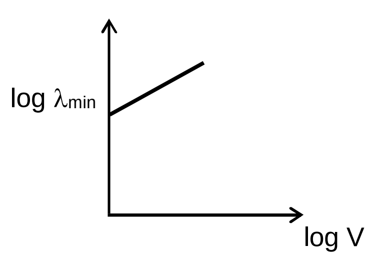
$= \frac{1}{2}\varepsilon_0 \frac{1}{(\mu_0\varepsilon_0)}(B_0)^2$ $\left[\because\ c^2 = \frac{1}{\mu_0\varepsilon_0}\right]$

$(U_{av})_{\text{electric field}} = \frac{1}{2\mu_0}B_0^2 = (U_{av})_{\text{magnetic field}}$

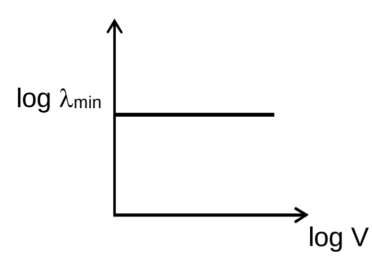
$\frac{(U_{ave})_{\text{electric field}}}{(U_{av})_{\text{magnetic field}}} = \frac{1}{1}$

5. An electron beam is accelerated by a potential difference V to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-ray. If λ_{min} is the smallest possible wavelenghth of X-ray in the spectrum, the variation of log λ_{min} with log V is correctly represented in

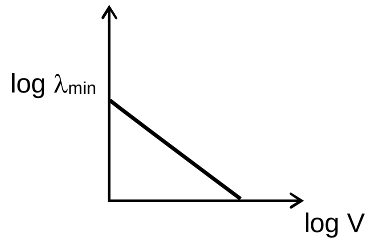
(A)



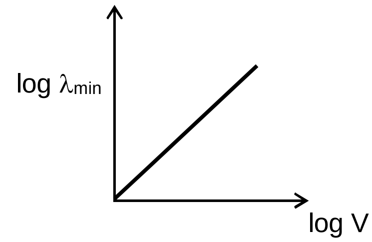
(B)



(C)



(D)



Ans. C

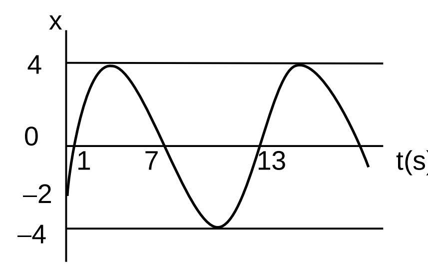
Sol. In X-ray tube

$$\lambda_{min} = \frac{hc}{eV}$$

$$\log\lambda_{min} = \log\left(\frac{hc}{e}\right) - \log V$$

Clearly, log λ_{min} versus log V graph slope is negative and intercept on y-axis is positive

6. The displacement (x) is plotted with time (t) for a particle executing SHM as shown below. The correct equation of SHM is



(A) $x = 4\sin\left(\frac{\pi t}{6} + \frac{\pi}{6}\right)$

(B) $x = 4\sin\left(\frac{\pi t}{6} - \frac{\pi}{6}\right)$

(C) $x = 2\sin\left(\frac{\pi t}{6} + \frac{\pi}{6}\right)$

(D) $x = 2\sin\left(\frac{\pi t}{6} - \frac{\pi}{6}\right)$

Ans. B

Sol. $x = 4\sin(\omega t + \phi)$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$x = 4\sin\left(\frac{\pi t}{6} + \phi\right)$$

$-2 = 4 \sin \phi$

$$-\frac{1}{2} = \sin\phi$$

$$\phi = -\frac{\pi}{6}$$

$$x = 4\sin\left(\frac{\pi t}{6} - \frac{\pi}{6}\right)$$

7. Consider single slit experiment of diffraction of light. If the light of wavelength 5000 A° fall on a slit of width 1 μm then the angular width of central maximum is

(A) 30°

(B) 15°

(C) 60°

(D) none of these

Ans. C

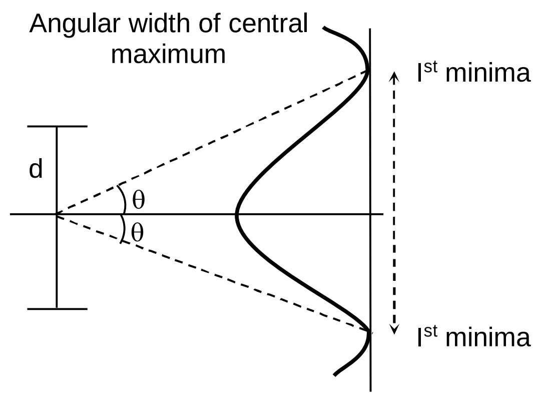
Sol. $d \sin \theta = \lambda$

$\sin\theta = \frac{\lambda}{d}$

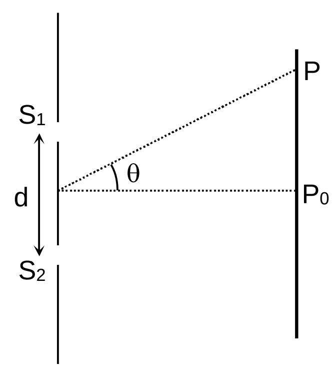
For first minima = $\frac{5\times10^{-7}}{10^{-6}}$

$\theta = 30°$ angular width of first minima

$2\theta = 60°$



8. Two identical sources each of intensity I_0 have a separation $d = \frac{\lambda}{8}$, where λ is the wavelength of the waves emitted by either source. The phase difference of the sources is $\frac{\pi}{4}$. The intensity distribution $I(\theta)$ in the radiation field as a funciton of θ, which specifies the direction from the soruces to the distant observation point P is given by



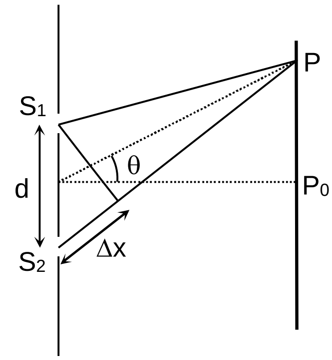
(A) $I(\theta) = 4I_0 \cos^2 \theta$

(B) $I(\theta) = \frac{I_0}{4}\cos^2\left(\frac{\pi\theta}{8}\right)$

(C) $I(\theta) = 4I_0 \cos^2\left[\frac{\pi}{8}(\sin\theta + 1)\right]$

(D) $I(\theta) = I_0 \sin^2\left[\frac{\pi}{8}(\sin\theta + 1)\right]$

Ans. C



Sol. $\Delta x = d\sin\theta = \frac{\lambda}{8}\sin\theta$

$$\Delta\phi = \frac{2\pi}{\lambda}\left(\frac{\lambda}{8}\sin\theta\right) + \frac{\pi}{4}$$

$$= \frac{\pi}{4}(1 + \sin\theta)$$

$$\because\ I(\theta) = 4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$I(\theta) = 4I_0\cos^2\left[\frac{\pi}{8}(1+\sin\theta)\right]$$

9. A closed pipe of length 22 cm, when excited by a 1875 Hz source forms standing waves. The number of pressure nodes formed in the pipe are (velocity of sound in air = 330 m/s)

(A) 1

(B) 2

(C) 4

(D) 3

Ans. D

Sol. Velocity of sound = 330 m/s
Length of closed pipe = 22 cm = 22×10^{-2} m
The fundamental frequency of stationary wave,

$$\nu_0 = \frac{\nu}{4\ell} = \frac{330}{4 \times 22 \times 10^{-2}} = \frac{3000}{8} = 375\text{ Hz}$$

$$3\nu_0 = 375 \times 3 = 1125\text{ Hz}$$

$$5\nu_0 = 375 \times 5 = 1875\text{ Hz}$$

Thus we get 5^{th} harmonic and number of nodes = 3
Note : In closed pipe, odd harmonics are produced.

10. If velocity, force and time are taken to be fundamental quantities, then the dimensional formula for mass is

(A) $Q = Kv^{-1}FT$

(B) $Q = Kv^{3}FT^{2}$

(C) $Q = 2Kv^{-2}FT$

(D) $Q = -3Kv^{-2}FT$

Ans. A

Sol. Let the quantity be Q , then
Q = f(v,F,T)
Assuming that the function is the product of power functions of v, F and T

$$Q = Kv^xF^yT^z \qquad \ldots(i)$$

Where K is a dimensionless constant of proportionality. The above equation dimensionally becomes

$\left[Q\right]=\left[LT^{-1}\right]^{x}\left[MLT^{-2}\right]^{y}\left[T\right]^{z}$

i.e., $\left[Q\right]=\left[M^{y}\right]\left[L^{x+y}T^{-x-2y+z}\right]$ …(ii)

Now,

Q = mass i.e., [Q] = [M]

So equation (ii) becomes

$\left[M\right]=\left[M^{y}L^{x+y}T^{-x-2y+z}\right]$

Its dimensional correctness require

y = 1, x + y = 0 and –x –2y +z = 0

Which on solving yields

x = –1, y = 1 and z = 1

substituting it in equation (i), we get

$Q = Kv^{-1}FT$

11. A 3.6 m long vertical pipe is filled completely with a liquid. A small hole is drilled at the base of the pipe due to which liquid starts leaking out. The pipe resonates with a tuning fork. The first two resonances occur when height of water column is 3.22 m and 2.34 m respectively. The area of the cross section of the pipe is

(A) 25π cm^2

(B) 100π cm^2

(C) 200π cm^2

(D) 400π cm^2

Ans. B

Sol. $f=\dfrac{(2n+1)v}{4(\ell+e)}$

$\ell_1+e=\dfrac{v}{4f}$ and $\ell_2+e=\dfrac{3v}{4f}$

$\Rightarrow \dfrac{\ell_2+e}{\ell_1+e}=3$

$\ell_2=(3.6-2.34)$m and $\ell_1=(3.6-3.22)$ m

$\Rightarrow e=0.06\text{ m}=0.6\text{ r}$

$\Rightarrow$ r = 0.1 m

$\Rightarrow$ A = 100 π cm^2

12. A stationary source of sound is emitting wave of frequency 30 Hz towards a stationary wall. There is an observer standing between the source and the wall. If the wind blows from the source to the wall with a speed 30 m/s, then the number of beats heard by the observer is (velocity of sound with respect to wind is 330 m/s)

(A) 10

(B) 3

(C) 6

(D) zero

Ans. D

Sol. Doppler's effect depends upon velocity of approach and separation of source and observer, hence no change in frequency received by the observer.
Hence no beat is heard.

13. A simple pendulum is suspended from the ceiling of an empty box falling near earth surface. The total mass of the system is M. The box experience air resistance $\vec{R} = -k\vec{v}$, where v is the velocity of the box and k is positive constant. After some time it is found that the period of oscillation of pendulum becomes double the value when it would have suspended from a point on earth. The velocity of box at that moment (take g in air same as on earth's surface)

(A) $\frac{Mg}{4k}$

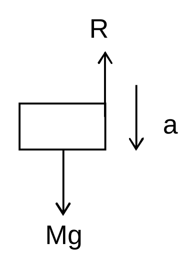
(B) $\frac{Mg}{k}$

(C) $\frac{Mg}{2k}$

(D) $\frac{2Mg}{k}$

Ans. A

Sol. $T = 2\pi\sqrt{\frac{\ell}{g-a}}$



Where a is the downward acceleration of box.

$T_0 = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow a = \frac{3g}{4}$ (as T = $2T_0$)

$\because$ For Box; Mg – R = Ma

$\Rightarrow$ R = Mg/4

and v = R/k

$\therefore\ v = \frac{Mg}{4k}$

14. A bird is flying up at an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with the horizontal. A fish in a pond look at the bird. When it is vertically above the fish. The angle at which the bird appears to fly (to the fish) is $\left[n_{water} = \frac{4}{3}\right]$

(A) $\sin^{-1}\left(\frac{3}{5}\right)$

(B) $\sin^{-1}\left(\frac{4}{5}\right)$

(C) 45°

(D) $\sin^{-1}\left(\frac{9}{16}\right)$

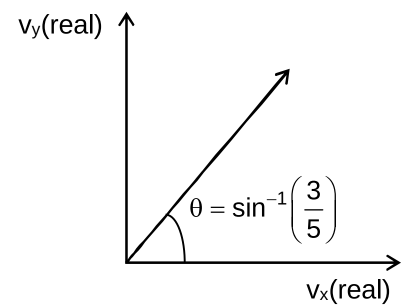
Ans. C

Sol. Let y-axis be vertically upwards and x-axis be horizontal

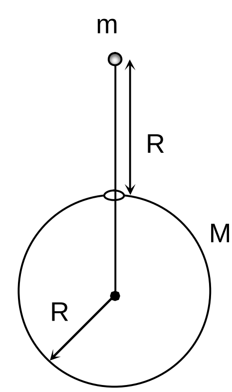
$$v_y(\text{app}) = \frac{v_y(\text{real})}{\left(\frac{1}{\mu}\right)}$$

$$v_x(\text{app}) = v_x(\text{real})$$

$$\tan\phi = \frac{v_y(\text{app})}{v_x(\text{app})} = \frac{4}{3}\tan\theta = \frac{4}{3}\times\frac{3}{4} = 1$$



15. A thin spherical shell of total mass M and radius R is held fixed. There is a small hole in the shell. A particle of mass m is released from rest at a distance R from the hole as shown in the figure. This particle subsequently moves under gravitational force of the shell. How long does it take to travel from the hole to the point diametrically opposite?



(A) $\sqrt{\frac{2R^3}{GM}}$

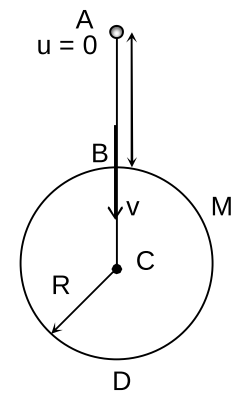
(B) $2\sqrt{\frac{2R^3}{GM}}$

(C) $2\sqrt{\frac{R^3}{GM}}$

(D) $3\sqrt{\frac{2R^3}{GM}}$

Ans. C

Sol. Let v be the velocity of the particle at point B. Applying conservation fo mechanical energy at point A and B.



$$-\frac{GMm}{2R} = -\frac{GMm}{R} + \frac{1}{2}mv_B^2$$

$$\Rightarrow v_B = \sqrt{\frac{GM}{R}}$$

Thereafter, $v_B = \sqrt{\frac{GM}{R}}$ = constant

(Hence, inside shell, $\vec{F}_g = m\vec{E}_g = 0$)

(Hence, $t_{BD} = \frac{2R}{v_B}$)

16. Refractive index of a prism is $\sqrt{\frac{7}{3}}$ and the angle of prism is 60°. The minimum angle of incidence of a ray that will be transmitted through the prism is

(A) 30°

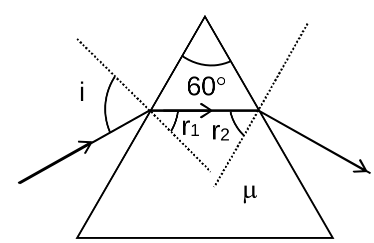
(B) 45°

(C) 15°

(D) 50°

Ans. A

Sol. $r_2 < \theta_C, \quad A - r_1 < \theta_C$



$\Rightarrow r_1 > A - \theta_C$

$\sin r_1 > \sin(A - \theta_C)$

$\frac{\sin i}{\mu} > \sin(A - \theta_C)$

$\Rightarrow \sin i > \mu \sin(A - \theta_C) > \mu[\sin A \cdot \cos\theta_C - \cos A \cdot \sin\theta_C]$

$\sin i > \frac{1}{2}$

$\Rightarrow i > 30°$

17. If potential energy between a proton and an electron is given by $|U| = \frac{ke^2}{2R^3}$, where e is the charge of electron and R is the radius of atom, then radius of Bohr's orbits given by (h = Plank's constant, K = constant)

(A) $\frac{ke^2m}{h^2}$

(B) $\frac{6\pi^2}{n^2}\frac{ke^2m}{h^2}$

(C) $\frac{2\pi}{n}\frac{ke^2m}{h^2}$

(D) $\frac{4\pi^2}{n^2}\frac{ke^2m}{h^2}$

Ans. B

Sol. $U = -\frac{Ke^2}{2R^3}$

$F = -\frac{dU}{dR} = -\frac{3Ke^2}{2R^4}$

But $F = \frac{mv^2}{R}$

$\Rightarrow \frac{mv^2}{R} = \frac{3ke^2}{2R^4}$ …(i)

Also, $mvR = \frac{nh}{2\pi}$ …(ii)

Solving (i) and (ii), we get

$R = \frac{6\pi^2}{n^2}\frac{ke^2m}{h^2}$

18. In an experiment, a small steel ball falls through a liquid at a constant speed of 10 cm/sec. If the steel ball is pulled upward with a force equal to twice its effective weight, how fast will it move upward?

(A) 5 cm/s

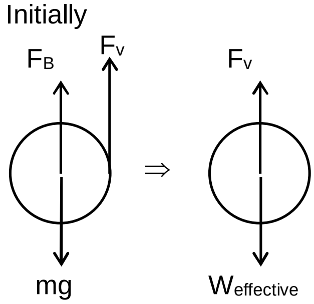
(B) zero

(C) 10 cm/s

(D) 20 cm/s

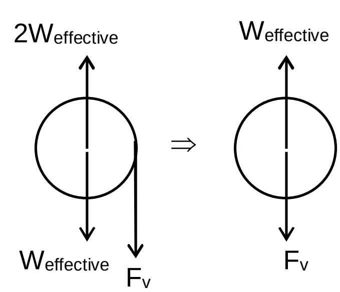
Ans. C

Sol. Initially



Where $W_{effective} = mg - F_B$

Finally



This is same situation as earlier but in oppsite direction

$\Rightarrow$ speed = 10 m/s in downward direction

19. AM radio band has carrier frequency from 500 kHz to 1500 kHz. Assuming a fixed inductance in a simple LC circuit, what is ratio of highest capacitance to lowest capacitance required to cover this frequency range?

(A) 3

(B) 6

(C) 9

(D) 12

Ans. C

Sol. $2\pi \times 500 = \omega = \dfrac{1}{\sqrt{LC_H}}$

$2\pi \times 1500 = \dfrac{1}{\sqrt{LC_L}}$

$\dfrac{1}{3} = \sqrt{\dfrac{C_L}{C_H}} \Rightarrow \dfrac{C_H}{C_L} = 9$

20. A particle of mass 4 kg executes SHM such that its energy of SHM is 36 J. Its velocioty at t = 0 is –3 m/s and displacment from the mean position is +20 cm. The equation of its motion is :

(A) $x = 20\sqrt{2}\sin\left(15t + \dfrac{3\pi}{4}\right)$ cm

(B) $x = 40\sqrt{2}\sin\left(20t + \dfrac{5\pi}{6}\right)$ cm

(C) $x = 40\sin\left(15t + \dfrac{2\pi}{3}\right)$ cm

(D) $x = 20\sqrt{2}\sin\left(20t + \dfrac{5\pi}{6}\right)$ cm

Ans. A

Sol. $\dfrac{1}{2}k \times A^2 = 36$

$v = \omega\sqrt{A^2 - x^2}$

$3 = \omega\sqrt{A^2 - 0.04}$

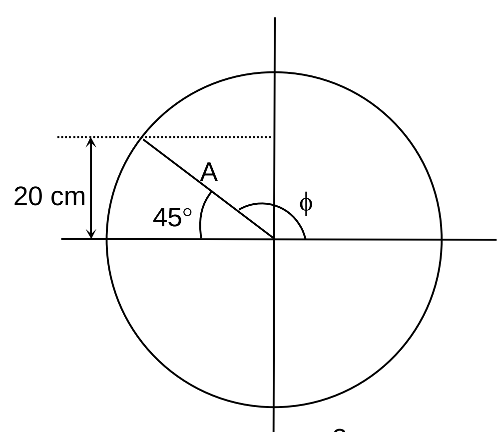
$9 = \dfrac{kA^2 - 0.04k}{m}$

36 = 72 – 0.04 k

0.04 k = 36 $\Rightarrow$ k = 900

$\omega = \sqrt{\dfrac{900}{4}} = \dfrac{30}{2} = 15$

$A = \sqrt{\dfrac{72}{900}} = \sqrt{\dfrac{8}{100}} = 20\sqrt{2}$ cm

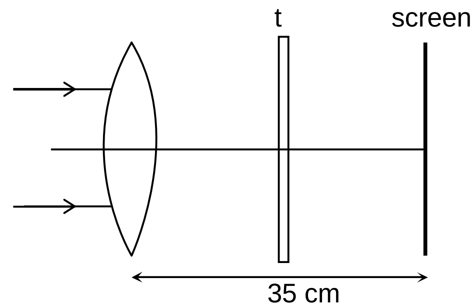


Initial phase $\dfrac{3\pi}{4}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. Parallel beam is incident on a thin lens (μ = 1.4) and radii 25 cm of each of the surface as shown. What should be the thickness of a slab (in cm) (μ = 1.5) between the lens and the screen so that final image is formed on the screen.



Ans. 00011.25

Sol. $\frac{1}{f} = 0.4 \times \frac{2}{25}$
$\Rightarrow$ f = 31.25
$NS = 3.75 = t\left(1 - \frac{1}{1.5}\right)$
$\Rightarrow$ t = 3.75 $\times$ 3 = 11.25 cm

22. A sample of β active nuclei has a half-life of 100 minutes. Initially there are $10^{13}\beta$ active nuclei. Assume that all β particles emitted leave the sample. What is the charge acquired (in μC) by the sample in 200 minutes?

Ans. 00001.20

Sol. $N = N_0 e^{-\lambda t}$ (where $\frac{N}{N_0} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$)
$N_{decays} = N_0\left(1 - e^{-\lambda t}\right) = 10^{13}\left(1 - \frac{1}{4}\right) = \frac{3}{4} \times 10^{13}$
$Q = 1.6 \times 10^{-19} \times \frac{3}{4} \times 10^{13} = 1.2\ \mu C$

23. In a coolidge tube, the potential difference used to accelerate the electron increased from 24.8 kV to 49.6 kV. As a result, the difference between the wavelength of K_α-line and minimum wavelength becomes two time. Then the wavelength (in A^o) of K_α-line is $\left[\frac{hc}{e} = 12.4 \text{ kV A}^o\right]$

Ans. 00000.75

Sol. Let ΔE be the energy of K_α-line then $\frac{hc}{\lambda_{k\alpha}} = \Delta E_{k\alpha}$

$\Rightarrow \lambda_{k\alpha} = \frac{hc}{\Delta E_{K\alpha}}$

Now, $\Rightarrow \lambda_{min} = \dfrac{hc}{e \times 24.8\ kV} = \dfrac{12.4}{24.8} A^o = \dfrac{1}{2} A^o$ and

$\Rightarrow \lambda_{2min} = \dfrac{hc}{e \times 49.6\ kV} = \dfrac{12.4}{49.6} A^o = \dfrac{1}{4} A^o$

$\Rightarrow 2\left(\lambda_{k\alpha} - \dfrac{1}{2}\right) = \left(\lambda_{k\alpha} - \dfrac{1}{4}\right)$

$\Rightarrow 2\lambda_{k\alpha} - 1 = \lambda_{k\alpha} - \dfrac{1}{4}$

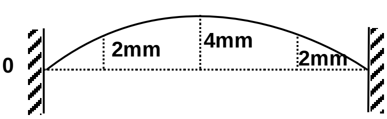
$\Rightarrow \lambda_{k\alpha} = 1 - \dfrac{1}{4}$

$\Rightarrow \lambda_{k\alpha} = 0.75\ A^o$

24. A string of length 1.5 m with its two ends clamped is vibrating in the fundamental mode. The amplitude at the centre of the string is 4 mm. Find the minimum distance (in meter) between the two points having amplitude of 2 mm.

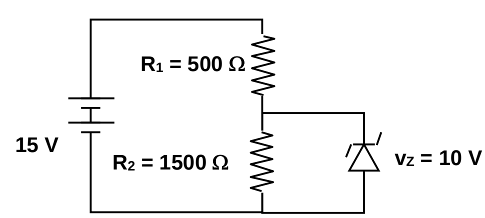
Ans. 00001.00

Sol. $\ell = \dfrac{\lambda}{2} \Rightarrow \lambda = 2\ell = 3$



Equation of stationary wave
y = [2A sin Kx] cos (ωt)
Amplitude = 2A sin Kx
$\Rightarrow 2 = 4\sin kx$
$\sin Kx = \dfrac{1}{2}; K = \dfrac{2\pi}{\lambda}$
$\therefore kx_1 = \dfrac{\pi}{6}, kx_2 = \dfrac{5\pi}{6}$
$\therefore x_2 - x_1 = \dfrac{5\pi}{6k} - \dfrac{\pi}{6k} = \dfrac{\lambda}{3} = 1\,m$

25. In the given figure, the current through zenor diode (in mA) is



Ans. 00003.33

Sol. $I_{R_2} = \dfrac{v_z}{1500} = \dfrac{1}{150} A$

$I_{R_1} = \dfrac{15 - v_z}{R_1} = \dfrac{1}{100} A$

$\Rightarrow I_z = I_{R_1} - I_{R_2} = \dfrac{1}{300} A = 3.33\ mA$

26. A compound microscope is used to enlarge an object kept at a distance of 3 cm from its objective. The objective consists of several convex lenses in contact and has a focal length of 2 cm. If a lens of focal length 10 cm is removed from the objective, the eyepiece has to be moved by x cm to refocus the image. The value of x is

Ans. 00009.00

Sol. I[st] case : $\frac{1}{v_1}-\frac{1}{(-3)}=\frac{1}{2} \Rightarrow v_1 = 6$ cm

When one lens is removed, the new focal length of the objective is

$$\frac{1}{F'}=\frac{1}{F}-\frac{1}{f_1}=\frac{1}{2}-\frac{1}{10}$$

$\Rightarrow F' = 2.5$ cm

The new postion of the image is

$$\frac{1}{v_2}-\frac{1}{(-3)}=\frac{1}{2.5} \Rightarrow v_2 = 15 \text{ cm}$$

The position of the image changes by 15 – 6 = 9 cm. Here, eye piece must be moved by the same distance (=9 cm) to refocus the image.

27. In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm and each rotation corresponds to one main scale division. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error. What is the thickness (in mm) of the wire?

Ans. 00002.15

Sol. Least count = $\frac{0.25}{5\times100}\text{cm} = 5\times10^{-4}\text{cm}$

Reading = 4×0.05 cm + $30 \times 5 \times 10^{-4}$ cm = 2.15 mm

28. Sound wave of frequency 600 Hz falls normally on a perfectly reflecting wall. The shortest distance (in cm) from the wall at which the air particles will have a maximum amplitude of vibration will be (speed of sound = 300 m/s)

Ans. 00012.50

Sol. The wall acts like rigid boundary and reflects this wave and sends it back towards the open end. At the open end antinode is formed and a node is formed at the wall.

The distance between antinode and node = $\lambda/4$

So, wavelength of note emitted = $\lambda = \frac{300}{600}$

$$\lambda = \frac{1}{2}\text{m}$$

So, maximum amplitude is obtained at a distance = $\lambda/4$

$$= \frac{1}{2}\times\frac{1}{4}=\frac{1}{8}\text{ m}$$

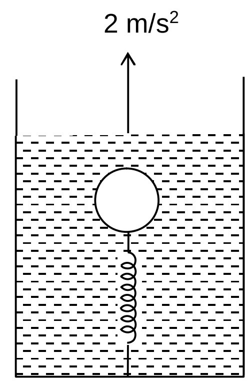
29. Assume pupil diameter to be 2.5 mm. Assume light of wavelength 500 nm. What is the minimum distance (in mm) between two point objects that can be seen clearly by the eye when kept at least distnace of distinct vision?

Ans. 00000.06

Sol. $\frac{x}{d} = \frac{1.22\lambda}{D}$

$$x = \frac{1.22 \times 5 \times 10^{-7} \times 0.25}{2.5 \times 10^{-3}} = 0.061\,\text{mm}$$

30. A ball of mass 10 kg and density 1 gm/cm^3 is attached to the base of a container having a liquid of density 1.1 gm/cm^3, with the help of a spring as shown in the figure. The container is going up with an acceleration 2 m/s^2. If the spring constant of the spring is 200 N/m, the elongation (in cm) in the spring is (There is no relative motion between ball and container) (g = 10 m/s^2)



Ans. 00006.00

Sol. The force acting on the ball are

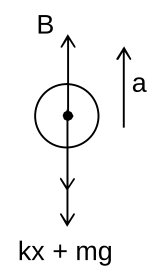
B – Mg – kx = Ma

$V_0 s_\ell (g + a) - V_0 sg - kx = V_0 sa$

$kx = V_0 s_\ell (g + a) - V_0 s(g + a)$

$= V_0 (g + a)(s_\ell - s) = \frac{M}{s}(12)(0.1) \times 10^3 = \frac{10}{10^3} \times 12 \times 10^2$

x = 6 cm



Chemistry PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which is incorrect?

(A) Iproniazid - Transquilizer

(B) Codein – non-narcotic

(C) Prontosil - Antibiotic

(D) Chloramphenicol – Broad spectrum

Ans. B

Sol. Codein is a narcotic drug.

32. Salts of metals W, X and Y are electrolysed under identical conditions using same quantity of electricity. If was observed that 4.8 g of W, 5.8 g of X and 19.2 g of Y were deposited at respective cathode. If the atomic weights of W, X and Y are 8, 29 and 64 respectively, then the ratio of their valencies is

(A) 2 : 3 : 1

(B) 1 : 2 : 3

(C) 1 : 3 : 2

(D) 3 : 2 : 2

Ans. C

Sol. $W_1 = \frac{AM_1}{nf_1} \qquad W_2 = \frac{AM_2}{nf_2} \qquad W_3 = \frac{AM_3}{nf_3}$

$nf_1 : nf_2 : nf_3 = 1 : 3 : 2$

33. Which of the following statement is incorrect?

(A) Black phosphorous is not toxic while white phosphorous is toxic.

(B) White phosphorous is not soluble in NaOH and does not evolve phosphine gas while red phosphorous is soluble in NaOH and evolves phosphine gas.

(C) Black phosphorous is the thermodynamically most stable allotrope among white phosphorous, red phosphorous and black phosphorous.

(D) White phosphorous is soluble in CS_2 while red phosphorous is not soluble in CS_2.

Ans. B

34. Which of the following statement is incorrect?

(A) SO_2 is used as an anti-chlor.

(B) S_2 molecule has two unpaired electrons in π^* (π antibonding) orbital.

(C) SO_3 gas is produced as by-product of the roasting of FeS_2.

(D) O_3 is dark blue in liquid state.

Ans. C

Sol. SO_2 gas is produced as by-product of the roasting of FeS_2.

35. Which of the following pair is paramagnetic?

(A) $\left[Co(ox)_3\right]^{3-}, \left[FeF_6\right]^{3-}$

(B) $\left[Co(ox)_3\right]^{3-}, \left[Fe(ox)_3\right]^{3-}$

(C) $\left[CoF_6\right]^{3-}, \left[Fe(CN)_6\right]^{4-}$

(D) $\left[Fe(CN)_6\right]^{3-}, \left[CoF_6\right]^{3-}$

Ans. D

Sol. $\left[Co(ox)_3\right]^{3-}$ and $\left[Fe(CN)_6\right]^{4-}$ do not have unpaired electrons while $\left[FeF_6\right]^{3-}$ and $\left[Fe(ox)_3\right]^{3-}$ have unpaired electrons and $\left[CoF_6\right]^{3-}$ has 4 unpaired electrons.

36. The calculated spin only magnetic moments of $\left[Ni(NH_3)_6\right]^{2+}$ and $\left[CrF_6\right]^{3-}$ in B.M. respectively, are (Atomic mass of Cr and Ni are 24, 28, respectively)

(A) 0, 3.87

(B) 3.87, 1.73

(C) 2.83, 3.87

(D) 1.73, 4.9

Ans. C

Sol. $\left[Ni(NH_3)_6\right]^{2+}$ has two unpaired electrons hence $\mu = \sqrt{8} = 2.83$ BM

$\left[CrF_6\right]^{3-}$ has three unpaired electrons hence $\mu = \sqrt{15} = 3.87$ BM

37. Among the following metal carbonyl species, the one with the weakest carbon-oxygen bonding is

(A) $Cr(CO)_6$

(B) $\left[Ti(CO)_6\right]^{2-}$

(C) $\left[V(CO)_6\right]^{-}$

(D) $\left[Mn(CO)_6\right]^{+}$

Ans. B

38. Which one of the following impurities present in colloidal solution cannot be removed by electrodialysis?

(A) Sodium sulphate

(B) Potassium chloride

(C) Glucose

(D) Barium nitrate

Ans. C

Sol. Glucose is non-electrolyte hence can't be removed by electrodialysis.

39. Which of the following statement is incorrect?

(A) Actinoid contraction is greater from element to element than lanthanoid contraction because of poor shielding of 5f orbitals than 4f orbitals.

(B) The magnetic properties of the actinoids are more complex than those of the lanthanoids.

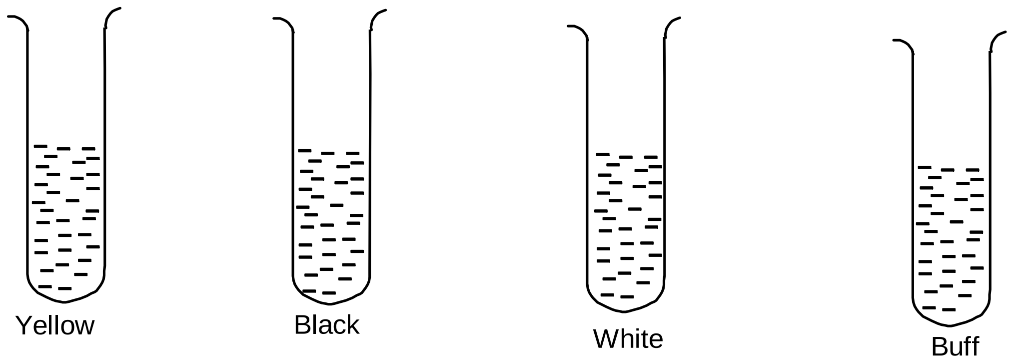
(C) The study of lanthanoids is easier than actinoids because actinoids are radioactive elements.

(D) Eu^{3+} is a strong oxidizing agent changing to Eu^{2+}, because Eu^{2+} has f^7 configuration.

Ans. D

Sol. Eu^{2+} is a strong reducing agent changing to the common +3 state.

40. H_2S gas passed in all the following test tube having one of the ions $Zn^{2+}, Ni^{2+}, Mn^{2+}$ and Cd^{2+} so that that precipitation is observed. Which is correct match



(A) Ni – yellow

(B) Cd - buff

(C) Zn - white

(D) Mn - black

Ans. C

Sol. NiS – black, CdS – yellow, Zn – white, Mns – buff.

41. Select the incorrect observation(s) for a 8.21 litre container, filled with 2 moles of Ar at 300 K

(A) It has pressure 6 atm if Ar behaves as an ideal gas and walls of container are rigid.

(B) If the container is opened at one side, moles of gas diffused from container when heated to 400 K is 1.75 if Ar behaves as an ideal gas.

(C) If Ar behaves as a non-ideal gas and volume of gas is negligible compared to the total volume occupied by gas, it would have pressure greater than 6 atm.

(D) If it is in closed non-rigid (like thin balloon) container and Ar gas behaves an ideal gas, its volume increases to 16.42 lit on heating to 600 K.

Ans. C

Sol. $$\left(P + \frac{an^2}{V^2}\right)V = nRT$$

$$P = \frac{nRT}{V} - \frac{an^2}{V^2}$$

Hence, would have pressure lesser than 6 atm.

42. Which of the following statement is incorrect?

(A) In the metallurgy of aluminium, purified Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity.

(B) In the metallurgy of gold by forest cyanide process Zn acts as a reducing agent.

(C) Van Arkel method is used for refining Zirconium.

(D) Pig iron obtained from blast furnace contains about 1% carbon and many impurities in smaller amount.

Ans. D

Sol. Pig iron contains about 4% carbon.

43. Which of the following statement is incorrect regarding complex $\left[Fe(H_2O)_5NO\right]^{2+}$?

(A) The oxidation state of iron in the complex is +2.

(B) Complex is formed during the brown ring test for nitrate ion.

(C) The colour is because of charge transfer.

(D) There are three unpaired electrons so that its magnetic moment is 3.87 BM.

Ans. A

Sol. NO transfer its electron to Fe^{2+} so that iron exists as Fe(I) and NO as NO^{+}.

44. Which of the following statement is correct?

(A) Photochemical smog occurs in a day time while classical smog occurs in the morning hours.

(B) Classical smog has an oxidizing character whereas the photochemical smog is reducing in character

(C) During formation of smog, the level of ozone in the atmosphere goes down.

(D) Classical smog is good for health but not photochemical smog.

Ans. C

Sol. Conceptual

45. Select the incorrect statement

(A) Silicon carbide is an example of network solid.

(B) The number of atoms present in one unit cell of body centred cubic is two.

(C) In face centred cubic unit cell two tetrahedral voids are present at each body diagonal of the cube.

(D) Coordination number of Na^{+} with respect to Na^{+} in NaCl is 6.

Ans. D

Sol. Coordination number of Na^{+} with respect to Na^{+} in NaCl lattice is 12.

46. Which of the following does not give precipitate with K_2CrO_4?

(A) Ag^{+}

(B) Pb^{2+}

(C) Ca^{2+}

(D) Sr^{2+}

Ans. C

47. 54 g of aluminium is treated with 100 ml of 0.4 M NaOH solution. The volume of H_2 gas evolved at 1 atm and 300 K is _____(assume that H_2 behaves as an ideal gas and R = 0.082 $L\ atm\ mol^{-1}K^{-1}$, atomic mass of Al = 27)

(A) 73.8 L

(B) 1.48 L

(C) 36.9 L

(D) 2.96 L

Ans. B

Sol. $2H_2O + 2NaOH + 2Al \longrightarrow 2NaAlO_2 + 3H_2$

Moles of $NaOH = \dfrac{100 \times 0.4}{1000} = 4 \times 10^{-2}$

Moles of Al $= \dfrac{54}{27} = 2$

Limiting reagent is NaOH hence moles of H_2 produced $= \dfrac{3}{2} \times 4 \times 10^{-2} = 6 \times 10^{-2}$

PV = nRT

$V = \dfrac{6 \times 10^{-2} \times 0.082 \times 300}{1}$

= 1.48 L

48. The enthalpy change for the following reactin is 369 kJ.

$NF_3(g) \longrightarrow N(g) + 3F(g)$

The average N – F bond energy is

(A) 369 $kJ\ mol^{-1}$

(B) 1107 $kJ\ mol^{-1}$

(C) 184.5 $kJ\ mol^{-1}$

(D) 123 $kJ\ mol^{-1}$

Ans. D

Sol. $\Delta_r H = (BE)_R - (BE)_P$

$369 = 3 \times (BE)_{N-F} - 0$

$(BE)_{N-F} = \frac{369}{3} = 123 \text{ kJ mol}^{-1}$

49. Which of the following statement is correct?

(A) Non-ideal solution obeys Raoult's law over the entire range of concentration.

(B) ΔH_{mix} and ΔV_{mix} for an ideal solution are less than zero.

(C) Non-ideal solution that shows positive deviation, the escaping tendency of each component decreases after mixing.

(D) Non-ideal solution that shows large negative deviation, form maximum boiling azeotropes.

Ans. D

50. Formula of metal oxide with metal deficiency defect in its crystal is $A_{0.7}O$. The crystal contains A^{2+} and A^{3+} ions. The fraction of metal existing as A^{2+} ions in the crystal is

(A) 0.96

(B) 0.14

(C) 0.50

(D) 0.31

Ans. B

Sol. $(x \times 2) + (0.7 - x) \times 3 = 2$

$2x + 2.1 - 3x = 2$

$0.1 = x$

$\text{Fraction} = \frac{0.1}{0.7} = \frac{1}{7} = 0.14$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The factor of ΔG values is important in metallurgy. The ΔG values for the following reactions at $800^{o}C$ are given as

$S_2(g)+2O_2(g)\longrightarrow 2SO_2(g)\quad \Delta G_1=-544\text{ kJ}$

$2Pb(s)+S_2(g)\longrightarrow 2PbS(s)\quad \Delta G_2=-287\text{ kJ}$

$2PbS(s)+O_2(g)\longrightarrow 2PbO(s)\quad \Delta G_3=-472\text{ kJ}$

The ΔG for the following reaction is – x kJ at 800°C. The X is______

$2PbS(s)+3O_2(g)\longrightarrow 2PbO(s)+2SO_2(g)$

Ans. 00729.00

Sol. $\Delta G=(\Delta G_1+\Delta G_3)-(\Delta G_2)$

$=(-544-472)-(-287)$

$=-1016+287$

$=-729\text{ kJ}$

52. Formation of polythene from calcium carbide takes place according to the following process

$CaC_2+2H_2O\xrightarrow[\text{yield}]{80\%}Ca(OH)_2+C_2H_2$

$C_2H_2+H_2\xrightarrow[\text{yield}]{70\%}C_2H_4$

$nC_2H_4\xrightarrow[\text{yield}]{100\%}\left(CH_2-CH_2\right)_n$

The amount of polyethylene (in g) obtained from 12.8 g of CaC_2 is_____

Ans. 00003.14

Sol. Moles of C_2H_4 formed $=0.2\times\frac{80}{100}\times\frac{70}{100}=0.112$

Amount of polyethylene formed $=0.112\times 28$

$= 3.136$

≈ 3.14 g

53. For the following cell reaction $A+2B^{+}\longrightarrow A^{2+}+2B$ the equilibrium constant at 300 K is 10^{15}. Find E^{o} for the $B^{+}+e^{-}\longrightarrow B$ at 300 K. (Given E^{o} for $A^{2+}+2e^{-}\longrightarrow A$ at 300 K is $+0.34$ V and $\frac{2.303R}{F}=0.0002$)

Ans. 00000.79

Sol. $E^{o}=\frac{2.303RT}{nF}\log K$

$E_C^{o}-E_A^{o}=\frac{0.0002\times 300}{2}\log 10^{15}$

$E^o_C = 0.01 \times 3 \times 15 + 0.34$
= 0.45 + 0.34
= 0.79 V

54. An LPG cylinder, containing 14 kg butane (assume it behaves as an ideal gas) at 27^oC and 20 atm pressure is leaking. After one day, its pressure reduced to 12 atm at the same temperature. The quantity of gas leaked (in kg) is ______

Ans. 00005.60

Sol. $P \propto W$

$\frac{P_1}{P_2} = \frac{W_1}{W_2}$

$\frac{20}{12} = \frac{14}{x}$

$x = \frac{12 \times 14}{20} = \frac{42}{5} = 8.4$ kg

Gas leaked = 14 – 8.4
= 5.6 kg

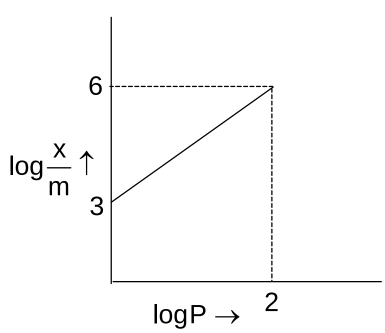
55. The tetrahedral voids present in 0.2 mole of a compound forming hexagonal close packing crystal structure are $x \times 10^{23}$. The x is_____(Given $N_A = 6 \times 10^{23}$)

Ans. 00002.40

Sol. Number of hexagonal units $= \frac{0.2 \times 6 \times 10^{23}}{6} = 0.2 \times 10^{23}$

Number of tetrahedral voids $= 0.2 \times 10^{23} \times 12$
$= 2.4 \times 10^{23}$

56. Adsorption of a gas follows Freundlich adsorption isotherm. x is the mass of the gas adsorbed on mass m of the adsorbent. The plot of $\log\frac{x}{m}$ versus logP is shown is the given graph.



$\frac{x}{m}$ is proportional to P^n, the n is_____

Ans. 00001.50

Sol. Slope $= \frac{3}{2} = 1.5$, hence $n = 1.5$.

57. Total number of isomers (including stereoisomers) of $\left[Co(ox)_3\right]^{3-}$ are______

Ans. 00002.00

58. In the conversion of PCl_5 to PCl_3

$PCl_5(g) \longrightarrow PCl_3(g) + Cl_2(g)$

the values of ΔH^o and ΔS^o are $180\ kJ\ mol^{-1}$ and $150\ J/K$ respectively at 300 K and 1 atm assuming that ΔH^o and ΔS^o do not change with temperature, then temperature (in K) above which conversion of PCl_5 to PCl_3 will be spontaneous is_____

Ans. 01200.00

Sol. $\Delta H^o = T\Delta S^o$

$$T = \frac{\Delta H^o}{\Delta S^o} = \frac{180 \times 1000}{150} = 1200\ K$$

59. Vapour pressure of $CHCl_3$ at 27^oC is 280 mm Hg. 2 g of non-volatile solute (molar mass $= 40\ g\ mol^{-1}$) is dissolved in 100 ml of $CHCl_3$. The vapour pressure (in mm Hg) of the solution is______(Given density of $CHCl_3$ $11.95\ g/mL$ and molar mass of H, C and Cl are 1, 12 and 35.5 $g\ mol^{-1}$ respectively)

Ans. 00278.60

Sol. Mass of $CHCl_3 = d \times V = 11.95 \times 100 = 1195\ g$

Moles of $CHCl_3 = \dfrac{1195}{119.5} = 10$ mole

Moles of non-volatile solute $= \dfrac{2}{40} = \dfrac{1}{20}$

$$\frac{P^o - P}{P^o} = \frac{n_B}{n_A} = \frac{\frac{1}{20}}{10}$$

$$\frac{280 - P}{280} = \frac{1}{200}$$

$$280 - P = \frac{280}{200} = 1.4$$

P = 280 -1.4

= 278.6 mm Hg

60. $wMg + xHNO_3 \longrightarrow yMg(NO_3)_2 + NH_4NO_3 + zH_2O$ (balanced equation) w, x, y and z are the stoichiometric coefficient of Mg, $HNO_3, Mg(NO_3)_2$ and H_2O respectively, the value of $\dfrac{x}{y} =$

Ans. 00002.50

Sol. $4Mg + 10HNO_3 \longrightarrow 4Mg(NO_3)_2 + NH_4NO_3 + 3H_2O$

$$\frac{x}{y} = \frac{10}{4} = 2.5$$

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The coefficient of x^{n-2} in the polynomial (x – 1)(x – 2)(x – 3) ….. (x – n) is

(A) $\frac{n(n^2+2)(3n+1)}{24}$

(B) $\frac{n(n^2-1)(3n+2)}{24}$

(C) $\frac{n(n^2+1)(3n+4)}{24}$

(D) $\frac{n(n^2+1)(3n-2)}{24}$

Ans. B

Sol. Coefficient x^{n-2} is sum of product of two natural numbers from first n natural numbers

$$= \frac{(1+2+3+.....+n)^2-(1^2+2^2+3^2+.....+n^2)}{2}$$

$$= \frac{1}{2}\left(\frac{n^2(n+1)^2}{4}-\frac{n(n+1)(2n+1)}{6}\right)$$

$$= \frac{1}{2}\cdot\frac{n(n+1)}{2}\left(\frac{n(n+1)}{2}-\frac{2n+1}{3}\right)$$

$$= \frac{n(n+1)}{4}\left(\frac{3n^2+3n-4n-2}{6}\right)$$

$$= \frac{n(n+1)(3n^2-n-2)}{24} = \frac{n(n^2-1)(3n+2)}{24}$$

62. If $\lambda = \sqrt{4+\sqrt{\frac{37b-81b^4-3}{b}}}$ where b > 0, then sum of all possible integral values of λ is

(A) 0

(B) 3

(C) 4

(D) 5

Ans. D

Sol. $\left(\lambda^2-4\right)^2=\dfrac{37b-81b^4-3}{b}=37-\left(81b^3+\dfrac{3}{b}\right)$

AM $\geq$ GM

$\dfrac{81b^3+\frac{1}{b}+\frac{1}{b}+\frac{1}{b}}{4}\geq\left(81b^3\cdot\dfrac{1}{b^3}\right)^{\frac{1}{4}}$

$81b^2+\dfrac{3}{b}\geq 4\times 3$; $81b^2+\dfrac{3}{b}\geq 12$

$81b^3+\dfrac{3}{b}=37-\left(\lambda^2-4\right)^2\geq 12$

$25 \geq (\lambda^2 - 4)^2$; $(\lambda^2 - 4)^2 \leq 5^2$

$(\lambda^2 - 9)(\lambda^2 + 1) \leq 0$; $\lambda^2 \leq 9$

$-3 \leq \lambda \leq 3$; $\lambda = -3, -2, -1, 0, 1, 2, 3$

$\lambda \neq -3, -2, -1, 0, 1$, so $\lambda = 2$ or 3

$\therefore$ Sum is 5

63. If $x^2y + y^2z + z^2x = 2022$; $x^2z + y^2x + z^2y = 2020$. Then value of $(x - y)^3 + (y - z)^3 + (z - x)^3$ is equal to

(A) –3

(B) –6

(C) –9

(D) none of these

Ans. B

Sol. $x^2y + y^2z + z^2x - x^2z - y^2x - z^2y = 2$

$x^2(y - z) + yz(y - z) - x(y^2 - z^2) = 2$

$(x - y)(y - z)(z - x) = -2$

So, $(x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x) = 3 \times -2 = -6$

64. If α, β, γ are the roots of the equation $x^3 - 3x^2 + 7x - 5 = 0$, then value of $\sum\alpha^2\beta$ is

(A) 4

(B) 6

(C) 8

(D) none of these

Ans. B

Sol. $\alpha + \beta + \gamma = 3$, $\alpha\beta + \beta\gamma + \gamma\alpha = 7$, $\alpha\beta\gamma = 5$

$(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) = 21$

$\alpha^2\beta + \alpha\beta^2 + \alpha\beta\gamma + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2 + \alpha^2\gamma + \alpha\beta\gamma + \gamma^2\alpha = 21$

$\sum\alpha^2\beta+3\alpha\beta\gamma=21$

$\sum\alpha^2\beta=21-3\alpha\beta\gamma=21-3\times 5=21-15=6$

$\sum\alpha^2\beta=6$

65. If imaginary part of $\frac{z-5}{e^{i\alpha}}+\frac{e^{i\alpha}}{z-5}$ is zero and if locus of z is a circle, then its radius is

(A) 1
(B) 4
(C) 9
(D) none of these

Ans. A

Sol. Let $w=\frac{z-5}{e^{i\alpha}}+\frac{e^{i\alpha}}{z-5}=t+\frac{\bar{t}}{|t|^2}$

Let $t=\frac{x-iy-5}{\cos\alpha+i\sin\alpha}$, $|t|=\sqrt{(x-5)^2+y^2}$

Im(w) = (y cos α – (x – 5) sin α)$\left(1-\frac{1}{(x-5)^2+y^2}\right)=0$

$(x-5)^2+y^2=1$, radius = 1

66. If $z^3+(3+2i)z+(-1+ia)=0$ has one real root, then the value of "a" can lie in the interval

(A) (–2, –1)
(B) (–1, 0)
(C) (0, 1)
(D) (1, 2)

Ans. B

Sol. Let z = α one real root $\alpha^3+(3+2i)\alpha+(-1+ia)=0$
$(\alpha^3+3\alpha-1)+i(2\alpha+a)=0$
So, $\alpha^3+3\alpha-1=0$ and $\alpha=-\frac{a}{2}$
$-\frac{a^3}{8}-\frac{3a}{2}-1=0$; $a^3+12a+8=0$
Let $f(a)=a^3+12a+8$, $f'(a)=3a^2+12>0$
So, f(a) is increasing has only one real root
$f(-1)<0$, $f(0)>0 \Rightarrow f(-1)f(0)<0$
So, $a\in(-1, 0)$

67. If $(1+x)^n=\sum_{r=0}^{n}a_r x^r$, $b_r=1+\frac{a_r}{a_{r-1}}$ and $\prod_{r=1}^{n}b_r=\frac{(101)^{100}}{100!}$, then value of n is

(A) 50
(B) 100
(C) 101
(D) none of these

Ans. B

Sol. $a_r = {}^nC_r$, $b_r = 1 + \frac{{}^nC_r}{{}^nC_{r-1}} = 1 + \frac{n-r+1}{r} = \frac{n+1}{r}$

$\prod_{r=1}^{n} b_r = \frac{n+1}{1} \cdot \frac{n+1}{2} \cdot \frac{n+1}{3} ..... \frac{n+1}{n} = \frac{(n+1)^n}{n!}$

So, n = 100

68. If $x + \frac{1}{x} = 1$, $p = x^{4000} + \frac{1}{x^{4000}}$ and q is the digit at unit place in the number $2^{2^n} + 1$, $n \in N$ and n > 1, then p + q is

(A) 8

(B) 6

(C) 7

(D) 5

Ans. B

Sol. $x^2 - x + 1 = 0$, $x = -w, -w^2$

$p = w^{4000} + \frac{1}{w^{4000}} = w + \frac{1}{w} = -1$

Since, n > 1, so $2^n = 4k$, $2^{4k} + 1 = (16)^k + 1$

So, unit place is 7 = q

p + q = –1 + 7 = 6

69. The number of triplets satisfying the equation $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 11$ where x, y, z are positive integers

(A) 0

(B) 3

(C) 6

(D) 12

Ans. B

Sol. $x^3 + y^3 + z^3 = 10$

(2, 1, 1), (1, 2, 1) (1, 1, 2), so three triplets

70. If $f(x) = a + bx + cx^2$ and α, β, γ are roots of $x^3 - 1 = 0$, then $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is equal to

(A) $f(\alpha)\, f(\beta)\, f(\gamma)$

(B) $3f(\alpha)\, f(\beta)\, f(\gamma)$

(C) $-f(\alpha)\, f(\beta)\, f(\gamma)$

(D) $-3f(\alpha)\, f(\beta)\, f(\gamma)$

Ans. C

Sol. $\alpha = 1, \beta = w, \omega^2 = \gamma$

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right) = -f(\alpha)\, f(\beta\, f(\gamma)$$

71. If $A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$, then the value of Det(AB + 2I) where B = adj A and I is identify matrix of order 3

(A) 64

(B) 125

(C) 216

(D) 343

Ans. C

Sol. $|A| = 4$, $|AB + 2I| = (|A| + 2)^3 = (4 + 2)^3 = 6^3 = 216$
Since, B = adj A so A adj A = $|A|I + 2I = (|A| + 2)I$
$\det((|A| + 2)I) = (|A| + 2)^3$

72. If matrix $A = [a_{ij}]_{5 \times 5}$ such that $a_{ij} = \begin{cases} 2 \;;\; \text{where } i = j \\ 0 \;;\; \text{when } i \neq j \end{cases}$, then value of det(adj(adj(adj A))) is equal to

(A) $(2)^{220}$

(B) $(2)^{320}$

(C) $(2)^{420}$

(D) none of these

Ans. B

Sol. $A = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$, $|A| = 2^5$ and det(adj(adj(adj A)))= $\left(|A|\right)^{(n-1)^3}$

$= \left(2^5\right)^{(4)^3} = \left(2^5\right)^{64} = (2)^{320}$

73. If the number of ways of selecting m cards out of unlimited number of cards bearing the number 2, 3, 4 so that they can not be used to write the number 423 is 93. Then m is equal to

(A) 4

(B) 5

(C) 6

(D) none of these

Ans. B

Sol. We can choose either 2, 3 or 3, 4 or 2, 4 or 2 only or 3 only or 4 only
So, $2^m + 2^m + 2^m - 3 = 93$
$3 \cdot 2^m = 96$, $2^m = 32$, $m = 5$

74. The number of polynomials of the form $x^3 + ax^2 + bx + c$ that are divisible by $x^2 + 1$ where a, b, c $\in$ {1, 2, 3, 4, 5, 6, ….. 1000} is

(A) 625

(B) 729

(C) 825

(D) 1000

Ans. D

Sol. $x^3 + ax^2 + bx + c = 0$ for $x = i, -i$
$x = i$; $i^3 + ai^2 + bi + c = 0$, $(b-1)i + (c-a) = 0$
$x = -i$; $-i^3 + ai^2 - bi + c = 0$ $(1-b)I + (c-a) = 0$
So, $b = 1$, $c = a$
a, c can take 1000 values

75. Let p_n denote the probability of getting n heads when a fair coin is tossed m times. If P_4, P_5, P_6 are in Arithmetic Progression, then the sum of all possible values of m is

(A) 14

(B) 21

(C) 28

(D) 35

Ans. B

Sol. $P_4 = {}^mC_4\left(\frac{1}{2}\right)^m$, $P_5 = {}^mC_5\left(\frac{1}{2}\right)^m$, $P_6 = {}^mC_6\left(\frac{1}{2}\right)^m$

$2 \times 5 = P_4 + P_6$

$2 \cdot {}^mC_5 = {}^mC_4 + {}^mC_6 \Rightarrow m = 7, 14$

So, sum of all possible values = 21

76. The mean and variance of a binomial variable x are 2 and 1 respectively. Then the probability that x takes values greater than 1 is

(A) $\frac{11}{16}$

(B) $\frac{13}{16}$

(C) $\frac{15}{16}$

(D) $\frac{7}{16}$

Ans. A

Sol. np = 2, npq = 1, $p = q = \frac{1}{2}$, n = 4, p(x = 2) + p(x = 3) + p(x = 4)

$= {}^4C_2\left(\frac{1}{2}\right)^4 + {}^4C_3\left(\frac{1}{2}\right)^4 + {}^4C_4\left(\frac{1}{2}\right)^4 = \frac{{}^4C_2 + {}^4C_3 + {}^4C_4}{16} = \frac{11}{16}$

77. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 3$ if the projection of $\vec{b}$ along $\vec{a}$ is equal to that of $\vec{c}$ along $\vec{a}$ and $\vec{b}$, $\vec{c}$ are perpendicular to each other, then the value of $|\vec{a} - \vec{b} + \vec{c}|$ is equal to

(A) $\sqrt{13}$

(B) $\sqrt{14}$

(C) $\sqrt{15}$

(D) $\sqrt{17}$

Ans. B

Sol. $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$, $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{\vec{c} \cdot \vec{a}}{|a|} \Rightarrow \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ and $\vec{b} \cdot \vec{c} = 0$

$|\vec{a} - \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} + 2\vec{a} \cdot \vec{c}$ = 1 + 4 + 9 = 14

$|\vec{a} - \vec{b} + \vec{c}| = \sqrt{14}$

78. The locus of a point equidistant from two given points whose position vectors are $\vec{a}$ and $\vec{b}$ is equal to

(A) $\left(\vec{r}-\frac{\vec{a}+\vec{b}}{2}\right)\cdot(\vec{a}+\vec{b})=0$

(B) $\left(\vec{r}-\frac{\vec{a}+\vec{b}}{2}\right)\cdot(\vec{a}-\vec{b})=0$

(C) $\left(\vec{r}-\frac{\vec{a}+\vec{b}}{2}\right)\cdot\vec{a}=0$

(D) $\left(\vec{r}-(\vec{a}+\vec{b})\right)\cdot\vec{b}=0$

Ans. B

Sol. $|\overrightarrow{PA}|=|\overrightarrow{PB}|$

$|\vec{r}-\vec{a}|^2=|\vec{r}-\vec{b}|^2$

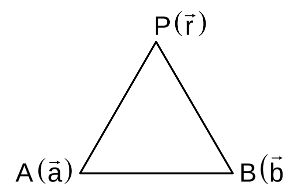
$|\vec{r}|^2+|\vec{a}|^2-2\vec{a}\cdot\vec{b}=|\vec{r}|^2+|\vec{b}|^2-2\vec{b}\cdot\vec{r}$

$|\vec{a}|^2-|\vec{b}|^2=2\vec{r}\cdot(\vec{a}-\vec{b})$

$2\vec{r}\cdot(\vec{a}-\vec{b})=|\vec{a}|^2-|\vec{b}|^2$

$2\vec{r}\cdot(\vec{a}-\vec{b})=(\vec{a}+\vec{b})(\vec{a}-\vec{b})$

$2\vec{r}-(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=0$



79. If the straight lines x = –1 + s, y = 3 – λs, z = 1 + λs and $x=\frac{t}{2}$, y = 1 + t, z = 2 – t with parameters s and t respectively are coplanar, then value of λ is

(A) –3

(B) –2

(C) 2

(D) none of these

Ans. B

Sol. $\frac{x+1}{1}=\frac{y-3}{-\lambda}=\frac{z-1}{\lambda}$ and $\frac{x-0}{1/2}=\frac{y-1}{1}=\frac{z-2}{-1}$ are coplanar

If shortest distance = $\begin{vmatrix}0+1 & 1-3 & 2-1\\ 1 & -\lambda & \lambda\\ \frac{1}{2} & 1 & -1\end{vmatrix}=0$, λ = –2

80. If a variable plane forms a tetrahedron of constant volume 64 units with coordinate planes, then the locus of the centroid of the tetrahedron is

(A) $x^2yz = 8$

(B) $xyz = 6$

(C) $xy^2z = 6$

(D) $xyz^2 = 6$

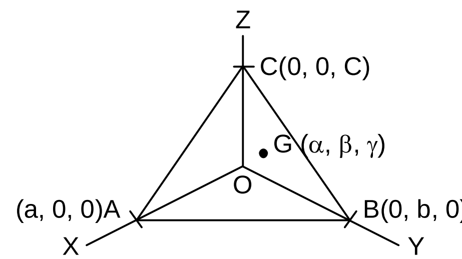
Ans. B

Sol. $\alpha = \frac{a}{4},\ \beta = \frac{b}{4},\ \gamma = \frac{c}{4}$

$a = 4\alpha$, $b = 4\beta$, $c = 4\gamma$

Volume of tetrahedron OABC = $\frac{1}{3}\left(\frac{1}{2}\times ab\right)\times c = \frac{abc}{6}$

$\frac{64\alpha\beta\gamma}{6} = 64$, $\alpha\beta\gamma = 6$, locus $xyz = 6$



SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $\left[\vec{a}\times(\vec{b}+\vec{c})\quad \vec{b}\times(\vec{c}-2\vec{a})\quad \vec{c}\times(\vec{a}+3\vec{b})\right] = k\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]^2$, then k is equal to

Ans. 00007.00

Sol. $\left[\vec{a}\times\vec{b}+\vec{a}\times\vec{c}\quad \vec{b}\times\vec{c}-2\vec{b}\times\vec{a}\quad \vec{c}\times\vec{a}+3\vec{c}\times\vec{b}\right]$

$= \left[\vec{a}\times\vec{b}-\vec{c}\times\vec{a}\quad \vec{b}\times\vec{c}+2\vec{a}\times\vec{b}\quad \vec{c}\times\vec{a}-3\vec{b}\times\vec{c}\right]$

Let $\vec{a}\times\vec{b}=\vec{p}$, $\vec{b}\times\vec{c}=\vec{q}$, $\vec{c}\times\vec{a}=\vec{r}$ = $[\vec{p}-\vec{r}\quad \vec{q}+2\vec{p}\quad \vec{r}-3\vec{q}]$

$= [\vec{p}-\vec{r}\quad 2\vec{p}+\vec{q}\quad -3\vec{q}+\vec{r}] = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & -3 & 1 \end{vmatrix}[\vec{p}\quad \vec{q}\quad \vec{r}]$

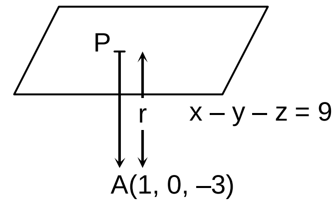
$= (1(1-0)-1(-6-0))\left[\vec{a}\times\vec{b}\quad \vec{b}\times\vec{c}\quad \vec{c}\times\vec{a}\right]$

$= 7\left[\vec{a}\times\vec{b}\quad \vec{b}\times\vec{c}\quad \vec{c}\times\vec{a}\right] = 7\left[\vec{a}\quad \vec{b}\quad \vec{c}\right]^2$

So, k = 7

82. The distance of the point (1, 0, –3) from the plane $x - y - z = 9$ measured parallel to the line $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$ is

Ans. 00007.00



Sol. Given plane is x – y – z = 9 ..... (1)

Given line is $\frac{x-2}{2}=\frac{y+2}{3}=\frac{z-6}{-6}$ ..... (2)

So, equation of line parallel to (2) and passing through points (1, 0, –3) is $\frac{x-1}{2}=\frac{y-0}{3}=\frac{z+3}{-6}=r$

x = 2r + 1, y = 3r, z = –3 – 6r

P(2r + 1, 3r, –6r – 3) where P lies on plane

So, 2r + 1 – 3r + 6r + 3 = 9 ; 5r = 5, r = 1

P(3, 3, –9) ; $AP=\sqrt{(3-1)^2+3^2+(-3+9)^2} = \sqrt{4+9+36}=\sqrt{49}=7$

So, AP = 7

83. If $0 \le \theta \le \pi$ and the system of equations x = (sin θ)y + (cos θ)z, y = z + x cos θ, and z = x sin θ + y has a non-trivial solutions, then $\frac{4\theta}{\pi}$ is equal to

Ans. 00003.00

Sol. $$\begin{vmatrix} 1 & -\sin\theta & -\cos\theta \\ \cos\theta & -1 & 1 \\ \sin\theta & 1 & -1 \end{vmatrix}=0$$

1(1 – 1) + sin θ(–cos θ – sin θ) – cos θ(cos θ + sin θ) = 0

–(sin θ + cos θ)^2 = 0, sin θ + cos θ = 0

sin θ = –cos θ, tan θ = –1, $\theta=\frac{3\pi}{4}, \frac{4\theta}{\pi}=3$

84. Let $A=\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$, and $(A + I)^{80} – 80A = \begin{bmatrix} a+1 & b+1 \\ c+1 & d-2 \end{bmatrix}$, then value of a + b + c + d is

Ans. 00001.00

Sol. $$A^2=\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}=0$$

So, $(I+A)^{80} = {}^{80}C_0 I + {}^{80}C_1 IA + ... + {}^{80}C_{80}A^{80}$

$(I+A)^{80}-80A=I$

$\begin{bmatrix} a+1 & b+1 \\ c+1 & d-2 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; a = 0, b = –1, c = –1, d = 3

a + b + c + d = 0 – 1 – 1 + 3 = 1

85. Let P(x) = 1 + x – x^2 – x^3 + x^4 + x^5 – x^6 – x^7 + ... ∞ where 0 < x < 1, if $P(x)=\frac{\sqrt{2}+1}{2}$, then value of (x + 1)^4 is equal to

Ans. 00004.00

Sol. p(x) = (1 + x) – x^2(1 + x) + x^4(1 + x) – x^6(1 + x) + ... ∞

P(x) = (1 + x)(1 – x^2 + x^4 – x^6 + – ∞) = (1 + x)$\left(\frac{1}{1+x^2}\right)$

$$P(x) = \frac{1+x}{1+x^2} = \frac{\sqrt{2}+1}{2}$$
$$(\sqrt{2}+1)(x^2+1) = 2x+2$$
$$(\sqrt{2}+1)x^2 + \sqrt{2}+1-2x-2 = 0$$
$$(\sqrt{2}+1)x^2 - 2x + (\sqrt{2}-1) = 0$$
$$x^2 - \frac{2}{\sqrt{2}+1}x + \frac{\sqrt{2}-1}{\sqrt{2}+1} = 0$$
$$x^2 - 2(\sqrt{2}-1)x + (\sqrt{2}-1)^2 = 0$$
$$\left(x-(\sqrt{2}-1)\right)^2 = 0 \ ; \ x = \sqrt{2}-1 \ ; \ x+1 = \sqrt{2}$$
$$(x+1)^4 = (\sqrt{2})^4 = 4$$

86. If x_1, x_2, x_3 are the roots of the equation $(2)^{33x-2} + (2)^{11x+2} = 2^{22x+1} + 1$, then $44(x_1 + x_2 + x_3)$ is equal to

Ans. 00008.00

Sol. Let $(2)^{11x} = t$
$$\frac{t^3}{4} + 4t = 2t^2 + 1$$
$t^3 + 16t = 8t^2 + 4$
$t^3 - 8t^2 + 16t - 4 = 0$ has roots t_1, t_2, t_3
$t_1\, t_2\, t_3 = 4$
$2^{11(x_1+x_2+x_3)} = 4$, $11(x_1 + x_2 + x_3) = 2$

87. Let α_r where r = 1, 2, 3, 4, ….., 100 be the roots of $\sum_{r=0}^{100}(z)^r = 0$. If $\sum_{r=1}^{100}\frac{1}{(\alpha_r - 1)} = 10\lambda$, then $|\lambda|$ is equal to

Ans. 00005.00

Sol. $1 + z + z^2 + ….. + 2^{100} = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3) ….. (z - \alpha_{100})$
$\log(1 + z + z^2 + ….. + z^{100}) = \log(z - \alpha_1) + \log(z - \alpha_2) + ….. + \log(z - \alpha_{100})$
Differentiating $\frac{1+2z+3z^2+.....+100z^{99}}{1+z+z^2+.....+z^{100}} = \frac{1}{z-\alpha_1} + \frac{1}{z-\alpha_2} + ... + \frac{1}{z-\alpha_{100}}$
Put z = 1
$$\frac{1+2+3+.....+100}{1+1+1+1+.....+1} = \frac{1}{1-\alpha_1} + \frac{1}{1-\alpha_2} + ... + \frac{1}{1-\alpha_{100}}$$
$$\frac{100\times 101}{2\times 101} = -\left(\frac{1}{\alpha_1-1} + \frac{1}{\alpha_2-1} + ... + \frac{1}{\alpha_{100}-1}\right)$$
$$-50 = \frac{1}{\alpha_1-1} + \frac{1}{\alpha_2-1} + ... + \frac{1}{\alpha_{100}-1}$$
$\sum_{r=1}^{100}\frac{1}{\alpha_r - 1} = -50 = 10\lambda$; $\lambda = -5$, so $|\lambda| = 5$

88. If $\left(x+\frac{1}{x}+1\right)^6 = a_0 + \left(a_1x+\frac{b_1}{x}\right)+\left(a_2x^2+\frac{b_2}{x^2}\right)+.....+\left(a_6x^6+\frac{b_6}{x^6}\right)$, then the sum of the digits of the value of a_0 is

Ans. 00006.00

Sol. $\left(1+x+\frac{1}{x}\right)^6 = {}^6C_0 + {}^6C_1\left(x+\frac{1}{x}\right)+{}^6C_2\left(x+\frac{1}{x}\right)^2+.....+{}^6C_6\left(x+\frac{1}{x}\right)^6 = \sum_{r=0}^{6}{}^6C_r\left(x+\frac{1}{x}\right)^r$

Constant term will develop if r is even integer r = 0, 2, 4, 6

$a_0 = {}^6C_0 + {}^6C_2 \cdot {}^2C_1 + {}^6C_4 \cdot {}^4C_2 + {}^6C_6 \cdot {}^6C_3$

$a_0 = 1+30+90+20 = 141$

$a_0 = 141$

89. The number of positive integer solutions of a + b + c = 60 where "a" is a factor of b and c is λ^2, then $\frac{\lambda}{4}$ is equal to

Ans. 00003.00

Sol. Since, "a" is a factor of b and c
So, a divides 60
So, a = 1, 2, 3, 4, 5, 6, 10, 12, 15, 30 ($a \neq 60$)
b = ma, c = na so a + ma + na = 60

a(1 + m + n) = 60, m + n = $\frac{60}{a}-1$, where m, n $\geq$ 1

So, solution is ${}^{\frac{60}{a}-1-1}C_{2-1} = \frac{60}{a}-2$

Total number of solution = 58 + 28 + 18 + 13 + 10 + 8 + 4 + 3 + 2 + 0 = 144

$\lambda^2 = 144$, λ = 12, $\frac{\lambda}{4} = 3$

90. The digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 are written in random order to form a nine digit number. Let p be the probability that this number is divisible by 36. Then the value of 36p is

Ans. 00008.00

Sol. Since, nine digit numbers consists of 1, 2, 3, ….., 9 where sum 1 + 2 + 3 + ….. + 9 = 45
So, it is divisible by 9, to be divisible by 36. This number should also be divisible by 4
So, last two digits must be 12, 16, 24, 28, 32, 36, 48, 52, 56, 64, 72, 76, 84, 92, 96
So there are 16 options. Number of favourable event = 16 $\times$ 7!

Total number of events = 9!, so probability $p = \frac{16\times 7!}{9\times 8\times 7!} = \frac{2}{9}$

9p = 2 $\Rightarrow$ 36p = 8